

A 28-nm 75-fs_{rms} Analog Fractional- N Sampling PLL With a Highly Linear DTC Incorporating Background DTC Gain Calibration and Reference Clock Duty Cycle Correction

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Abstract—An analog fractional- N sampling phase-locked loop (PLL) is presented. It achieves 75-fs rms jitter, integrated from 10 kHz to 10 MHz, and a -249.7 -dB figure of merit (FoM) at the fractional- N mode with a 52-MHz reference clock. The measured fractional spur is less than -64 dBc across the 5.5–7.3-GHz output frequency band. The PLL employs digital-to-time converter (DTC)-based sampling PLL architecture, high linearity DTC design techniques, background DTC gain calibration, and reference clock duty cycle correction (DCC) to improve the integrated phase noise (IPN) and fractional spur. This design meets the performance requirement of the 5G cellular 64-quadrature-amplitude modulation (QAM) standard in the 28-/39-GHz band, supporting 2×2 multi-in multi-out (MIMO). This paper, implemented in a 28-nm CMOS process, is integrated in a 5G millimeter-wave cellular transceiver. This PLL consumes 18.9 mW and occupies 0.45 mm².

Index Terms—Analog phase-locked loop (PLL), digital-to-time convert (DTC), DTC gain calibration, duty cycle correction (DCC), fractional- N PLL, low jitter, reference clock doubler, sampling phase detector (SPD), voltage-controlled oscillator (VCO).

I. INTRODUCTION

THE potential of 5G new radio (NR) to deliver higher data rates enhances user mobile experience, and also creates exciting opportunities in the development of new technologies, e.g., vehicle to everything (V2X) and machine-type communications (MTC). The 3GPP 5G standards demand stringent phase noise (PN) and spur requirements on the local oscillator (LO), as the LO PN defines the error vector magnitude (EVM) floor in a wireless transceiver once other signal impairments are calibrated [1]. The 5G NR utilizes both the sub-6-GHz and millimeter-wave frequency bands, i.e.,

26.5–29.5 and 37–40 GHz. For cellular operation at millimeter-wave frequencies, an integrated PN (IPN) of about -36 dBc is required to support 64-quadrature amplitude modulation (QAM) and 2×2 multiple-input and multiple-output (MIMO) under non-ideal channel conditions. At the highest frequency channel, i.e., 40 GHz, this requirement translates to 89-fs rms jitter, which is much more demanding than that required for 4G LTE supporting 256-QAM and 4×4 MIMO and WIFI 802.11ac standard specification (i.e., ~ 150 fs_{rms}) [2], [8].

Recent publications on digital-to-time converter (DTC)-based fractional- N phase-locked loops (PLLs) [3]–[14] have demonstrated a good IPN and a better figure of merit (FoM) compared to alternative fractional- N PLLs. The DTC cancels the quantization error (QE) due to the $\Sigma\Delta$ -modulator ($\Sigma\Delta$ M) for fractional- N synthesis so that a near-zero phase error (e.g., a few picoseconds) is presented at the phase detector (PD) input after locking. As a result, a high-gain PD can be used to suppress the noise contribution from the following stages [e.g., charge pump (CP) and loop filter (LF)]. DTC gain calibration is crucial for PLL performance. A digital PLL implementation facilitates the DTC calibration by employing a time-to-digital converter (TDC) to convert the timing error to a digital phase error [8]–[10]. However, the high-resolution TDC increases design complexity and power consumption. Alternatively, a binary PD was used in [3] and [12] for simplicity. However, bang-bang PLL has two main design challenges for use in wireless applications: quantization noise (QN) must be kept below the tolerable spot PN, and fast lock must be guaranteed even for wide frequency steps [12]. In addition, the design of a low PN, digitally controlled oscillator (DCO) becomes challenging at higher oscillation frequency, as the digitally controlled fine-tuning bank contributes additional routing parasitics and QN. Therefore, an analog PLL form is more preferred to achieve lower IPN and better FoM.

In this paper, we proposed a DTC-based sampling analog PLL architecture and design techniques [13] to break the performance barrier of traditional fractional- N PLLs. First, a two-

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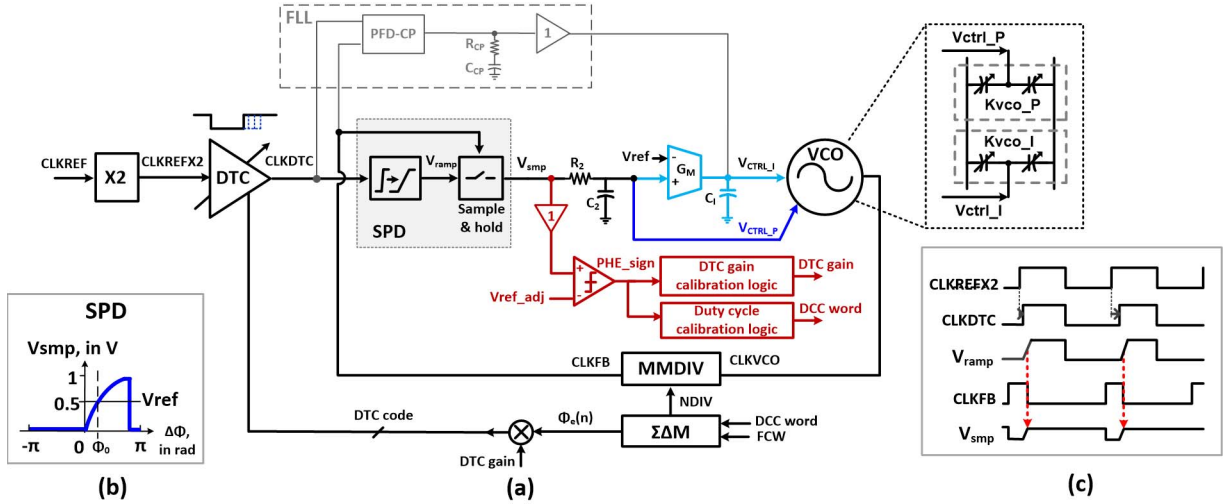


Fig. 1. (a) DTC-based fractional-*N* PLL with a reference clock doubler. (b) Characteristic of the SPD. (c) PLL timing diagrams.

stage sampling PD (SPD), followed by a resistor-less (LF, is introduced to tune the voltage-controlled oscillator (VCO) for low-power and low jitter requirements. Second, a single comparator with dynamically adjusted threshold enables robust and accurate DTC gain calibration in an analog PLL. Third, a reference clock doubler with background duty cycle correction (DCC) further improves the IPN. Using these techniques, the 5.5–7.3-GHz PLL achieves rms jitter as low as 75 fs and the state-of-the-art FoM of -249.7 dB with a 52-MHz crystal reference. It is further cascaded by a frequency multiplier to generate the LOs for a 28-GHz transceiver. The second-stage frequency multiplier can be designed to have very low IPN [15]–[17], resulting in a minor increase in the overall rms jitter. This design meets the requirements of the 5G cellular 64-QAM standards in the 28-/39-GHz band. It also satisfies the IPN requirements of the legacy 4G LTE standards and WIFI 802.11ax standard.

This paper is organized as follows. Section II provides an overview of the PLL architecture. Section III describes the DTC gain calibration. Section IV discusses the reference clock doubler and the DCC. Section V elaborates on the high linearity DTC and other circuit implementation, while Section VI presents the experimental results.

II. ARCHITECTURE OVERVIEW

Fig. 1 shows the system diagram of the proposed high-performance DTC-based sampling analog PLL. It contains an SPD and a multi-modulus feedback divider (MMDIV) driven by a ΣΔM for fractional-*N* synthesis. The expected phase error, Φ_e , is generated from the ΣΔM QE. After scaling by the estimated DTC gain, K_{DTC} , the resulting value modulates a DTC in the reference path to cancel the QE in CLKFB, which is similar to [8]. As a result, the required PD linear operation range after locking is reduced from a few VCO periods to only a few picoseconds.

In contrast to [7]–[11], which used a high-resolution multi-bit TDC to digitize the phase error, this design employs an SPD and keeps the signal path in analog to avoid the QN associated with the TDC. The SPD consists of a ramp

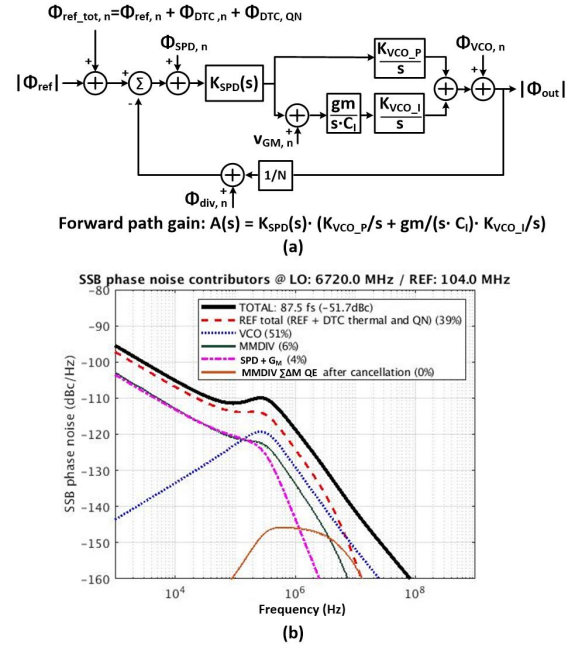


Fig. 2. (a) Analog sampling PLL linear model. (b) PLL PN spectrum according to the linear model.

generator and a sample and hold circuit. The rising edge of CLKDTC triggers a well-defined voltage ramp with a high dV/dt slope, (e.g., $K_{slope} = 5$ GV/s), which is then sampled by the falling edge of CLKFB, converting the phase error into a sampled voltage, V_{smp} , as illustrated in Fig. 1(c). After a first-order RC filter to remove the high-frequency noise and to suppress reference spur, the sampled voltage output then drives two paths: one path directly tunes the VCO (i.e., $V_{CTRL,P}$), providing proportional gain, $K_{VCO,P}$, and the other path goes through a voltage-to-current (G_M) circuit and an integration capacitor, C_I , to generate $V_{CTRL,I}$ for the VCO.

A linear phase-domain model for the proposed sampling PLL is shown in Fig. 2(a). Here, it is modeled as a continuous-time system, which is valid as long as the PLL

bandwidth (BW) is an order of magnitude smaller than that of the reference frequency, f_{ref} . The gain factor of the SPD, K_{SPD} , can be calculated as

$$K_{\text{SPD}} = \Delta v / \Delta \phi = K_{\text{slope}} / (2\pi f_{\text{ref}}). \quad (1)$$

For $f_{\text{ref}} = 104$ MHz and $K_{\text{slope}} = 5$ GV/s, $K_{\text{SPD}} = 48/2\pi$, which is more than ten times higher compared to the phase detection gain of a conventional PLL using a phase frequency detector (PFD) and a CP. Consequently, the G_{M} noise, $v_{\text{GM},n}$, is largely suppressed when referred to the input. Thus, G_{M} can be implemented with only a few microamperes. A smaller G_{M} also helps to reduce C_1 required to stabilize the loop for a given BW.

The proportional–integral (PI) filter configuration, as shown in Fig. 1, is a natural choice for an SPD due to several benefits. First, it eliminates the noisy resistor in a conventional passive analog LF. Second, according to the linear model in Fig. 2(a), the feed-forward path provides a zero at $(K_{\text{VCO}_I}/K_{\text{VCO}_P}) \cdot (G_{\text{M}}/C_1)/2\pi$. Therefore, for a given G_{M} , C_1 can be reduced by using a smaller $K_{\text{VCO}_I}/K_{\text{VCO}_P}$ ratio, resulting in a more area efficient LF implementation. Third, PLL BW is mainly determined by the SPD gain, K_{SPD} , and the VCO tuning sensitivity in the proportional path, K_{VCO_P} , and is insensitive to G_{M} , unlike in [5] and [6]. This reduces the loop BW variation to process, voltage, and temperature (PVT) variations, and also relaxes the gain accuracy requirement of the G_{M} circuit. There is no need to align the P and I paths intentionally. When the PLL is locked, V_{CTRL_P} equals V_{ref} plus the G_{M} input referred offset voltage to output a zero-averaged current to C_1 . The G_{M} offset voltage (i.e., typically <50 mV) is small enough to not affect K_{VCO_P} . Note that the voltage ripple at V_{smp} due to sampling can result in reference spur via both P and I paths. Although V_{CTRL_P} modulates the VCO directly in the P path, it results in a much lower level of reference spur compared to the integral path, as $K_{\text{VCO}_P}/K_{\text{VCO}_I}$ is $<1/10$ for the desired PLL BW (i.e., 600 kHz). An additional RC low-pass filter can be added after C_1 to further reduce the reference spur with little impact on the phase margin of the loop.

To strive for low rms jitter (i.e., 89 fs), we propose to use a reference clock doubler, as shown in Fig. 1. By doubling the PLL sampling rate, the feedback division ratio, N , in Fig. 2(a) is halved, thus, the in-band PN due to DTC, SPD, and G_{M} is reduced, resulting in the overall IPN improvement [20]–[23]. Fig. 2(b) shows the simulated PLL output PN spectrum with a clock doubler. The total noise contribution from the SPD, G_{M} , and MMDIV is kept as low as 10% of the overall IPN (integrated from 10 kHz to 200 MHz), so that the reference path (including the DTC) and the VCO dominate the PLL IPN. According to the linear analysis, the PLL IPN is improved by ~ 2 dB with the reference clock doubler.

Although using a sub-SPD (SSPD) as in [5], [6], and [18] can eliminate the MMDIV noise completely, we prefer to keep an MMDIV in the feedback path to gain the extra design freedom in K_{SPD} at a fractional cost of IPN [i.e., 6% in Fig. 2(b)]. The proposed SPD consists of a ramp generator to provide precise control and programmability of K_{SPD} . The use of the MMDIV reduces the ramp generator's operating frequency to f_{ref} , so that it can be implemented using a simple

RC charging circuit to achieve a wide programmable range. A smaller K_{SPD} provides a wider linear operation range for SPD and is preferred during the phase/frequency acquisition, while a larger K_{SPD} is desirable in the steady state to suppress the G_{M} noise at the cost of increased C_1 . As a tradeoff, K_{slope} of 5 GV/s is chosen in this design to keep $C_1 < 50$ pF in order to save chip area. K_{slope} can be calibrated to reduce the process variation, if needed. On the contrary, an SSPD provides a much higher gain, which is sensitive to PVT and is difficult to calibrate. K_{SSPD} is also calculated from (1). Assuming a 6-GHz sinusoidal input of 0.4-V peak swing, A_{VCO} , $K_{\text{slope}} = A_{\text{VCO}} \cdot 2\pi \cdot f_{\text{VCO}} = 15$ GV/s, which is three times of the slope of the SPD. Moreover, to achieve low reference spur (e.g., -70 dBc), isolation buffers are usually needed between the VCO and the SSPD to suppress the sampler kickback, which is a major cause of reference spur in an SSPLL. This isolation buffer further increases the signal slew rate at sampler input, and the resultant K_{SSPD} suffers from large PVT variations that are difficult to calibrate [5].

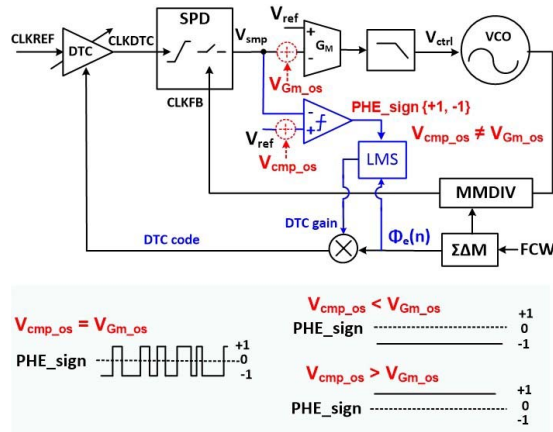
As indicated in Fig. 1, two digital calibration techniques are employed to assist the sampling analog PLL, which are necessary to achieve the state-of-the-art IPN and low spurious tones. A single comparator-based DTC gain calibration technique overcomes the calibration difficulties that existed in an analog PLL without employing a multi-bit TDC as in [7]–[11]. The reference clock DCC technique reuses the same comparator output to extract the duty cycle error and compensates it via the $\Sigma\Delta\text{M}$ and MMDIV in the feedback loop without additional analog circuitries. Both calibrations run in the background to ensure performance over PVT variations.

In addition, a PFD-CP-based frequency-locked loop (FLL) is used to speed up the initial phase/frequency acquisition. The SPD suffers from limited linear operation range, as shown in Fig. 1(b). Consequently, the sampling loop may take up to hundreds of microseconds to lock according to system simulation, depending on the initial phase and frequency error in the loop, the DTC gain initial error, and the reference clock duty cycle error. With the aid of an FLL, the simulated frequency acquisition time is reduced to within $5 \mu\text{s}$. The FLL is powered down after frequency locking to reduce power consumption. Since the noise of the PFD-CP is not a concern, implemented FLL occupies small extra silicon area.

III. DTC GAIN CALIBRATION

The basic form of the DTC gain calibration is a least-mean squares (LMS) regression, as described in [3] for a digital PLL.

However, it is not easy to apply it to an analog PLL. As shown in Fig. 3, PLL adjusts the locking phase so that the time-averaged output current of G_{M} is zero. Ideally, $V_{\text{smp}} = V_{\text{ref}}$ after the loop is locked, and thus, a voltage comparator with a threshold of V_{ref} can be used to obtain the polarity of the phase error, PHE_{sign} , for DTC calibration purposes. However, both G_{M} and the comparator have offsets (i.e., V_{Gm_os} and V_{cmp_os} as indicated in Fig. 3). As $V_{\text{Gm}_\text{os}} \neq V_{\text{cmp}_\text{os}}$, the comparator would output mostly “+1” or “−1” depending on which offset is larger. The output is dominated by the threshold mismatch between G_{M} and the comparator,

Fig. 3. Comparator and G_M offset and the impact on K_{DTC} calibration.

which corrupts any useful information for DTC calibration. As a result, the DTC gain, K_{DTC} , would deviate from the correct value and result in significant PN and spur degradation. The system simulation shows that a few millivolts threshold mismatch is sufficient to disturb the K_{DTC} adaptation loop. With a 10-mV threshold mismatch, the IPN is already severely degraded due to the significantly stronger fractional spur.

Since V_{smp} contains the G_M offset, the design in [5] and [6] extracted the sign of the G_M output current instead. However, it is cumbersome to extract the sign of a current (I), compared to that of a voltage (V). Additional circuitries were added in G_M to steer its current to a sign extracting node within the given reference period, converting I to V by charging/discharging the node, and then, the settled voltage is compared to the previous voltage hold by an unity-gain buffer to obtain the polarity of I [6]. The comparator offset remains an issue. To remove the offset, the raw PHE_sign was scaled by μ , and then fed to two accumulators connected in series to generate an overflow signal, which overrides the comparator output periodically. The choice of μ is tricky as it determines the overflow rate [6]. If the comparator offset is small, the override injections are triggered infrequently, they appear as low-level noise, and if the comparator offset is large, they are triggered frequently and may corrupt any useful information for calibration.

To overcome the aforementioned threshold voltage mismatch and the cumbersome task of detecting the sign of the G_M output current, we propose to adjust the comparator threshold voltage dynamically to track the G_M threshold voltage, rather than to rely on a pre-defined comparator threshold. As shown in Fig. 4, a comparator V_{ref} generation circuit adjusts V_{ref} to make the comparator output a zero-averaged bit sequence for the LMS-based calibration loop. When the PLL is locked, the comparator would have the same threshold voltage as the G_M circuit.

Three different implementations of the V_{ref} generation circuit are compared in Fig. 5. In Fig. 5(a), the comparator output is simply delayed by a 1-bit digital delay cell, and then controls a ΔV -digital-to-analog converter (DAC). In contrast to a regular DAC, the ΔV -DAC output voltage increases

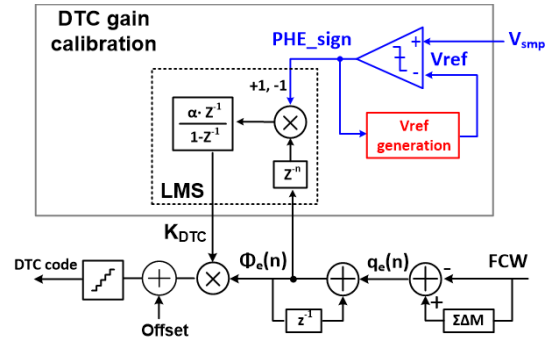
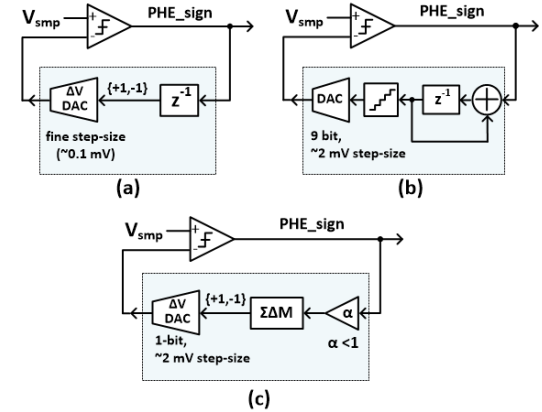


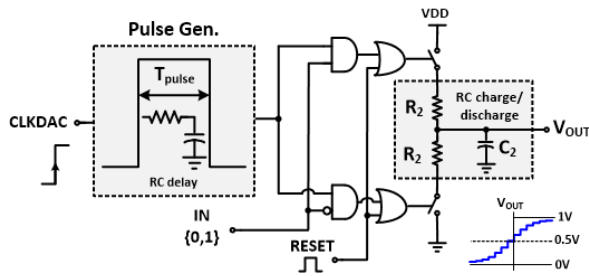
Fig. 4. Comparator with dynamically adjusted threshold for DTC gain calibration.

Fig. 5. Comparator V_{ref} generation circuits.

or decreases by a step size, ΔV , according to the input is “+1” or “-1,” respectively. Since there is no digital filtering, the toggling of the comparator output due to noise is directly transferred to the ΔV -DAC. As a result, the step size of ΔV -DAC needs to be very fine to avoid the V_{ref} variation due to noise degrading system performance. Based on the system simulation, a step size of ~ 0.1 mV is required. Several consecutive +1’s or -1’s are required to introduce a more significant change in V_{ref} .

To relax the DAC resolution, a digital accumulator followed by a regular DAC can be employed, as shown in Fig. 5(b). A digital accumulator implements a low-pass filter, which filters the comparator output toggling due to noise. As a result, the resolution of the DAC can be relaxed to 2 mV according to system simulation. The digital accumulator might have more bits than the DAC input, and a quantizer is used to generate the DAC code. A 9-bit DAC is required in order to generate V_{ref} of ~ 0.5 V to compare with the sampled voltage, V_{smp} .

To further simplify the DAC design, a $\Sigma\Delta M$ followed by a 1-bit ΔV -DAC is proposed in Fig. 5(c). The scaled comparator output is fed to a digital $\Sigma\Delta M$ to obtain the dc component of the input data and then used to adjust the comparator V_{ref} via ΔV -DAC. The $\Sigma\Delta M$ shaped the QN to higher frequencies, and then filtered by the integrating DAC. Although higher order can be used, a first-order $\Sigma\Delta M$ is sufficient to relax ΔV -DAC to 1 bit with a resolution of 2 mV. This method is employed in this design for simplicity.

Fig. 6. Schematic of ΔV -DAC.

Compared to the prior art, the threshold mismatch between G_M and the comparator is compensated via a 1-bit ΔV -DAC elegantly, rather than calibrating G_M and the comparator offset, respectively, as in [5] and [6].

The schematic of the 1-bit ΔV -DAC is shown in Fig. 6. It has two modes: reset and charge/discharge. During the reset mode, the DAC output is precharged to half-VDD via resistor division. Then, it discharges/charges by one step each clock cycle based on the 1-bit input from the $\Sigma\Delta M$ (“0” is discharging and “1” is charging). The voltage step size, ΔV , is proportional to the pulsewidth, T_{pulse} , as shown in (2)

$$\Delta V = \frac{\Delta Q}{C_2} \cong \begin{cases} T_{\text{Pulse}} \frac{V_{\text{DD}} - V_{\text{OUT}}}{R_2 \cdot C_2}, & \text{IN} = 1 \\ -T_{\text{Pulse}} \frac{V_{\text{OUT}}}{R_2 \cdot C_2}, & \text{IN} = 0, \end{cases} \quad (2)$$

As T_{pulse} is proportional to RC delay, ΔV depends on the RC ratio instead of the absolute R and C values, which are insensitive to process variation. Moreover, ΔV does not need to be accurate (~ 2 mV is acceptable) or linear for this purpose as long as the ΔV -DAC output is monotonic. Therefore, a simple RC ramp-based circuit is used. Compared to a conventional DAC, the proposed ΔV -DAC minimizes the chip area and design complexity.

A strong-arm latch-based dynamic comparator [19] is used for phase polarity detection. A unity-gain buffer is added prior to the comparator to reduce the comparator kickback, which could disturb the sampler output and lead to higher reference spur. Dynamically adjusting the comparator threshold restores a zero-mean input (i.e., PHE_sign) to the calibration circuit, which ensures fast and accurate DTC calibration by using 1-bit phase polarity data even with large threshold mismatch (verified up to 200 mV in simulation). The calibration time is less than 50 μs and the calibration error is within 1%. The simulated ΔV -DAC output voltage and K_{DTC} transient are shown in Fig. 9, which will be discussed together with the DCC loop in Section IV.

IV. REFERENCE CLOCK DUTY CYCLE CORRECTION

Reference clock doubler is introduced in Fig. 1 to further reduce the rms jitter. As shown in Fig. 7, the duty cycle error in CLKREF introduces even/odd period mismatch in the doubled clock, CLKREFX2, resulting in reference spur. Moreover, for a DTC-based sampling PLL, the duty cycle error can degrade PLL performance severely as it introduces an additional phase error between CLKREFX2 and CLKFB,

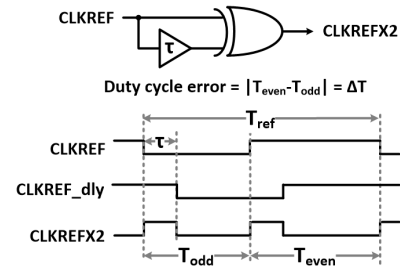


Fig. 7. Schematic of the CLKREF doubler.

which could exceed the linear operation range of the SPD (typically < 50 ps). For 1% duty cycle error in a 52-MHz CLKREF, the resultant period difference between even and odd cycles is 192 ps, which leads to a timing error of ± 96 ps between CLKREFX2 and CLKFB, in alternate cycles. Consequently, the loop dynamics are disturbed. In addition, the duty cycle error overwhelms the comparator output (i.e., PHE_sign) and disrupts or even fails the DTC calibration. Therefore, the utilization of a reference clock doubler in a DTC-based PLL has not been reported yet.

A DCC technique is proposed in Fig. 8 to calibrate and compensate the duty cycle error in background without additional analog circuitries. DTC gain calibration loop is also shown in Fig. 8(a), as both calibrations share the same comparator for PHE_sign. The principle of the DCC is also based on an LMS regression, which is the same as that used in a digital PLL [20]. When CLKREF has duty cycle error, the phase error detected by the SPD will alternate between positive and negative values, depending on the cycle status (e.g., “+1” for even cycle and “−1” for odd cycle), as shown in Fig. 8(b). This strong correlation is explored to form the LMS-regression loop in Fig. 8(a). In contrast to [20], which used a high-resolution TDC to digitize the phase error, this design reuses the 1-bit phase polarity data obtained for the DTC gain calibration for DCC as well.

To ensure fast convergence in the presence of DTC gain error, the comparator output is first differentiated ($1 - Z^{-1}$) to obtain the frequency error (freq_err) before correlating with the cycle status (even_cycle). The cycle status can be obtained by sampling the non-doubled reference clock using the falling edge of CLKFB [see Fig. 8(b)]. The calibration result (dcc_comp) is applied to the $\Sigma\Delta M$ to pull-in or push-back the phase of CLKFB alternately by the same amount (i.e., $\Delta T/2$) to compensate for the even/odd mismatch in CLKREFX2 [see Fig. 8(c)]. As a result, the SPD will not see the duty cycle error and remains in linear operation. The direction of the phase adjustment is controlled by the polarity of even_cycle. Since even and odd cycles always come in pair, the net dcc_comp input to the $\Sigma\Delta M$ is zero in each CLKREF. Therefore, dcc_comp only alters the phase of CLKFB without affecting its frequency, which is set by the frequency control word (FCW).

Compared to the other DCC techniques reported in [20]–[23], this design senses the CLKREF duty cycle error from the PD output via a single comparator, instead of

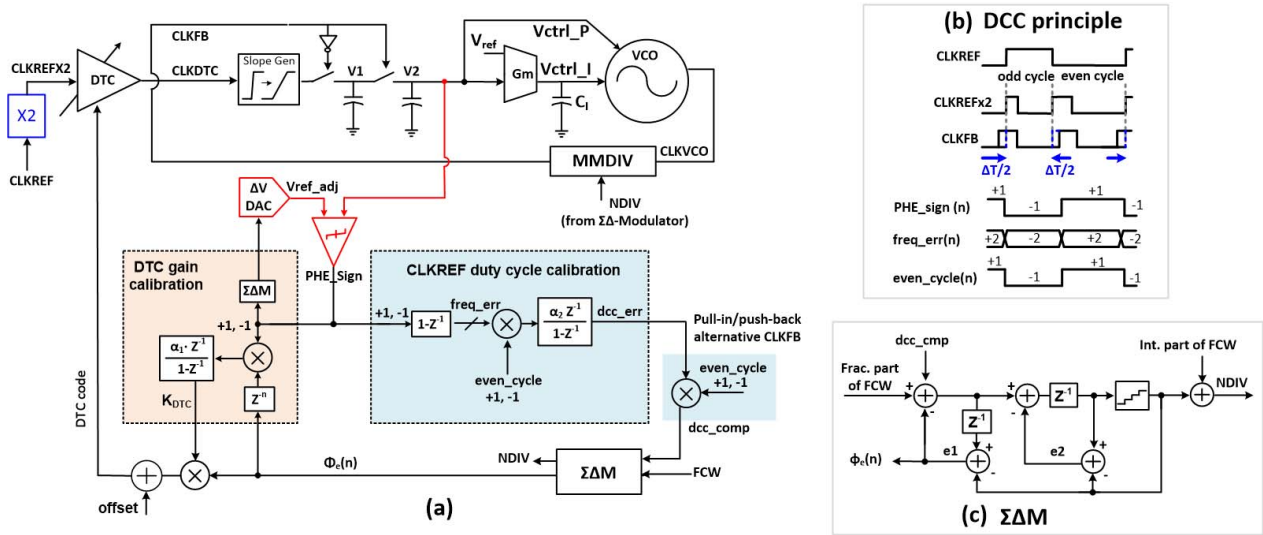


Fig. 8. (a) Single comparator-based DCC and K_{DTC} calibration in the analog fractional- N sampling PLL. (b) Operating principle of the DCC. (c) Implementation of the $\Sigma\Delta M$.

detecting from the doubled clock directly. The compensation via MMDIV avoids the additional noise contribution due to the delay line used in [21] and [22]. The noise due to the DCC loop is maintained 10 dB below the thermal noise of the CLKREF path. This is done by reducing the adaptation loop BW after convergence via decreasing the gain factor α_2 in Fig. 8(a). The DCC range of a few percent of T_{ref} is sufficient for most of the reference sources [e.g., crystal oscillator and temperature compensated crystal oscillator (TCXO)] and it can be easily extend to $>10\%$ of T_{ref} by increasing the integer bits of the fixed-point dcc_comp , if needed. Therefore, a higher DCC accuracy and larger correction range are both obtained with much less analog circuitry compared to the prior art.

Both calibration loops elaborated in Fig. 8(a) (i.e., DTC gain and DCC) are essential for the proposed sampling analog PLL to achieve high performance. Note that, both calibrations use the same 1-bit comparator output to minimize the hardware cost. We rely on the digital correlation and linear regression to extract K_{DTC} and duty cycle error, respectively, so that both calibrations run in the background, simultaneously. Ensuring that the PHE_sign is zero in average is crucial for both calibrations, as the threshold offset mismatch between G_M and the comparator might dominate the polarity of the phase error, obsoleting the duty cycle error information. These two calibrations can be enabled together from the beginning of the locking process, without a dedicated calibration sequence or time slot. Since the complexity of the digital calibration circuitry (see Fig. 8) is low, it consumes tens of microamperes and can run continuously in the background to track temperature and voltage variations. To reduce the settling time of the two calibration loops, a larger adaptation BW is employed initially after channel frequency change [controlled by α_1 and α_2 in Fig. 8(a)], and then gearshift to a smaller BW (i.e., $\alpha_1/4$ and $\alpha_2/4$) after 50 μs (controlled by a programmable timer) to reduce noise.

Fig. 9(a) shows the convergence trajectory of the DCC and the DTC gain for initial K_{DTC} of 200, 400, 800, 1200,

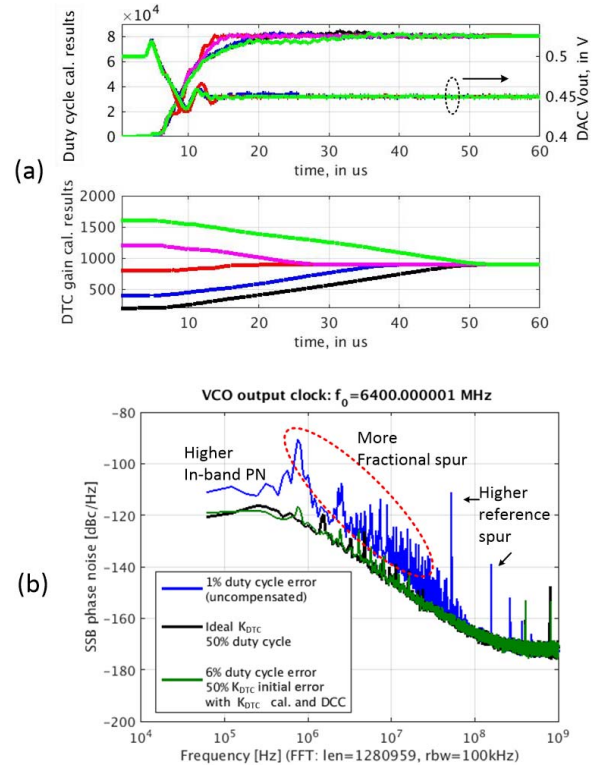


Fig. 9. (a) Simulated K_{DTC} and DCC transient for various initial K_{DTC} . (b) Simulated PLL PN spectrum with and without DCC.

and 1600, respectively (idea $K_{DTC} = 895$ in this case). The duty cycle calibration converges within 20 μs even with large initial K_{DTC} error. The dynamically adjusted comparator threshold settles to 0.45 V to compensate for the 50-mV G_M and comparator threshold mismatch in this case. The DTC gain calibration also converges once the duty cycle error is corrected. With the aforementioned gearshift technique, the worst case settling time for K_{DTC} is within 50 μs even for initial K_{DTC} error as large as $\pm 80\%$. As K_{DTC} and dcc_comp

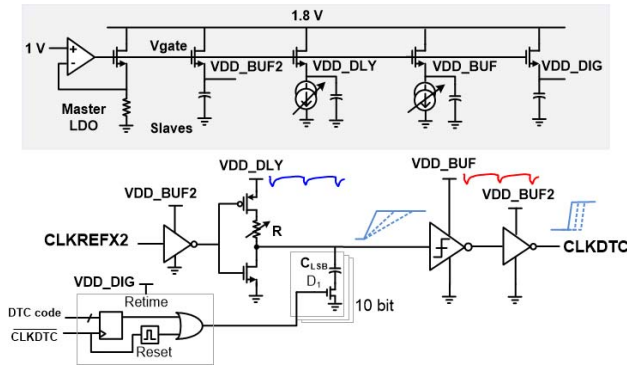


Fig. 10. Schematic of the DTC.

are both proportional to a VCO period, a lookup table of initial values can be built for different channel frequencies to reduce the settling time further, if needed.

Fig. 9(b) shows the comparison of the simulated PLL PN spectrum with and without DCC. Without compensation, PLL in-band PN degrades more than 10 dB, and significant amount of fractional spur and higher reference spur is observed with 1% duty cycle error, compared to a 50% duty cycle case. More degradation is observed when the duty cycle error increases to 6%. With the proposed DCC, even with 6% duty cycle error and large K_{DTC} initial error, the simulated PLL PN is similar to the ideal case. Both DCC and K_{DTC} calibrations operate in the background, without degrading PN or spurious tones.

The robustness of the proposed calibration techniques is also verified for the special FCWs in simulation, including integer- N , close to integer- N , and the cases when the fractional part of the FCW is 0.5 and 0.25. Thanks to the additional dithering due to the dcc_comp input to the $\Sigma\Delta M$, K_{DTC} always converges to <1% error for any FCWs.

V. CIRCUIT IMPLEMENTATION

A. Digital-to-Time Converter

DTC rms jitter contributes to the PLL in-band PN. The DTC nonlinearity increases fractional spur and noise folding [3]–[8]. This design uses an RC delay-based DTC topology for simplicity, as shown in Fig. 10. The DTC resolution is ~ 650 fs to keep the QN 10 dB below the thermal noise. The 10-bit delay is sufficient to support the VCO frequency of 5–7 GHz for a second-order $\Sigma\Delta M$. Resistor (R) programmability covers process variation, similar to [8]. The 10-bit fine-tuning implemented by a switched-capacitor array is controlled by the $\Sigma\Delta M$ sequence to cancel the QE. An inverter buffer following the delay stage restores the steep slope for the SPD. Although the DTC topology is simple, it is not trivial to achieve a PN floor below -160 dBc/Hz and an integral nonlinearity (INL) of ~ 1 LSB [6], [8], [25], [26]. Several design techniques are introduced in this design to meet the low PN and high linearity requirements, which will be elaborated in this section.

The unit delay capacitor, C_{LSB} , and the inverter buffer are both sized for the PN requirement. The DTC PN is code dependent (worst at the maximum code). For a given f_{ref} and resolution ($T_{res} = \ln 2 \cdot RC_{LSB}$), the single-sideband (SSB)

PN sampled at the half-VDD transition considering noise folding is approximated in (3), in which k_{slew} is the slew rate when output charges to half-VDD ($k_{slew} = (1/2RC)$), and ENBW is the effective noise BW of the RC delay stage ($ENBW = (1/2\pi RC) \cdot (\pi/2)$)

$$\mathcal{L} \cong 10 \cdot \log_{10} \left[\frac{1}{2} \left(\frac{2\pi \cdot f_{ref}}{k_{slew}} \right)^2 \cdot 4kTR \cdot \frac{ENBW}{f_{ref}/2} \right] \quad (3)$$

$$\mathcal{L} \cong 10 \cdot \log_{10} \left[2kT \cdot f_{ref} \left(\frac{2\pi}{\ln 2} \right)^2 \cdot \frac{2^n \cdot T_{res}^2}{C_{LSB}} \right]. \quad (4)$$

The PN at the mid-code is simplified to (4), which defines the minimum C_{LSB} required to achieve the PN floor target for a given T_{res} , f_{ref} and number of bits ($n = 10$ in this design). As the PN floor is dominated by thermal noise, flicker noise is not considered in (3) and (4) for simplicity. According to (4), we size the C_{LSB} to 2 fF so that the delay stage results in a PN of ~ -171 dBc/Hz to not dominate the DTC PN. Instead, the inverter buffer is dominant due to its slow input slope. The inverter buffer is sized up to a few hundreds of micrometer in width to sufficiently reduce the amount of flicker and thermal noise converting to the output clock jitter.

The slope-dependent (i.e., code dependent) propagation delay due to the inverter buffer is the main cause of the nonlinearity in an RC delay-based DTC [6]. It typically results in DTC INL. The DTC capacitive array has been sized with intrinsic matching for 10-bit accuracy, and thus, the simulated differential nonlinearity (DNL) is small (i.e., ± 0.2 LSB). To improve the INL, a fixed capacitor can be paralleled to the tuning capacitor bank to reduce the input slope variation over code. Since an overly sized inverter buffer is anyway required for PN, it also provides the desirable fixed capacitive loading (e.g., ~ 2 pF in this design) to keep the INL to ~ 1 LSB.

In addition to the static distortion, the code-dependent supply settling error also contributes to INL. DTC draws large transient current at each rising edge of CLKREFX2. Consequently, its supply voltage dips, and then resettles back each cycle, as shown in Fig. 10. This introduces code-dependent INL and the memory effect as the DTC delay for the current code also depends on the previous cycles' DTC codes due to the not fully settled supply. To speed up the supply settling, this design uses a master-slave source follower regulator (see Fig. 10) instead of a closed-loop regulation scheme that has limited BW. The delay stage, the first inverter buffer, and the digital control circuit each have its own slave branch to reduce supply-induced spur. Each slave branch consists of three NMOS source followers connected in series to regulate the 1.8 down to 1.0 V (only one stage is shown in Fig. 10 for simplicity), providing ~ 70 -dB supply rejection at low frequencies. The NMOS source follower is biased in sub-threshold, in which the drain current is exponential to the gate-source voltage. Therefore, the supply ripple due to the large transient load current variation is much smaller compared to when it is in saturation. The supply regulator contributes $\sim 25\%$ of the overall DTC PN at 100-kHz frequency offset and it reduces to 10% at 1-MHz offset where thermal noise dominates. Unlike in [25], R is in series with the PMOS to use the RC charging that starts from the ground, rather than the discharging that

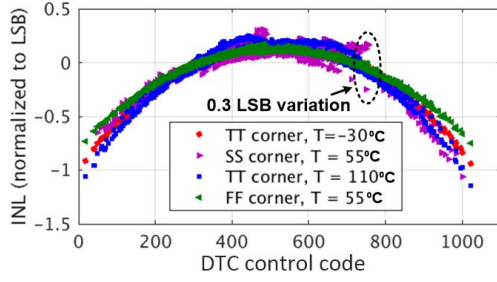
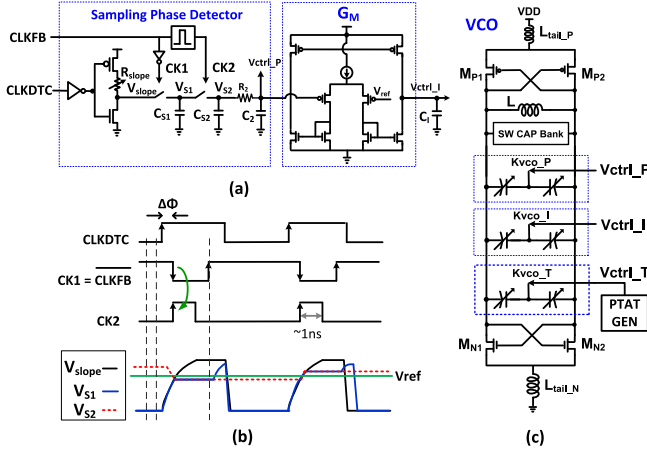


Fig. 11. Simulated DTC INL with MASH1-1 input.

Fig. 12. (a) Schematic of the SPD and G_M . (b) SPD timing diagram. (c) Schematic of the 6-GHz VCO.

depends on the supply. The DTC codes are reset to “1” each cycle to fully discharge the tuning capacitors before applying the new code (see Fig. 10). This helps to further reduce the code-dependence. Moreover, a programmable bleeding current is used for both the delay stage and the following inverter buffer. More dc bleeding current speeds up the settling of the source follower as the settling time is defined by gm/C (i.e., gm of the source follower and the bypass capacitor, C), at the cost of higher power consumption.

To capture the INL due to both static and dynamic distortion, DTC is simulated with actual $\Sigma\Delta M$ -sequence, and its INL curve is shown in Fig. 11 for different process corners at various temperatures. With 400- μA total bleeding current, the INL is close to the static sweep results (~ 1 LSB) and the spread for a given code due to memory effect is < 0.3 LSB, which is insignificant, and thus, the static distortion dominates. The maximum INL reduces from 2.4 to 1.2 LSB when I_{bleed} increases from 10 to 400 μA according to the circuit simulation. The bleed current can be switched off to save power when fractional spur and IPN requirements are relaxed.

B. Sampling Phase Detector, G_M , and VCO

The schematic of the SPD and G_M is shown in Fig. 12(a). It is beneficial to use an RC charging for ramp generation, similar to the DTC delay stage. Thus, the same resistor control code can be applied to DTC and SPD to account for the

process variation. Although the slope is nonlinear due to the exponential RC charging characteristic, the nonlinearity is negligible compared to DTC, as $\Delta\Phi$ only varies by a few picoseconds after locking with the aid of the K_{DTC} calibration and the DCC.

For the sampling stage, a two-stage sampler controlled by non-overlapping clock, CK1 and CK2, generates V_{S2} with a zero-order-hold nature, which eases the timing requirement of the G_M circuit and the dynamic comparator used for calibrations [see Fig. 12(b)]. In addition, it provides a first-order discrete-time infinite impulse response (IIR) filtering on the phase error to suppress the high-frequency noise. The IIR filter transfer function in the z -domain is

$$\frac{V_{S2}}{V_{slope}}(z) = \frac{\lambda \cdot z^{-0.5}}{1 - (1 - \lambda) \cdot z^{-1}}, \quad \text{in which } \lambda = \frac{C_{S1}}{C_{S1} + C_{S2}}. \quad (5)$$

Its equivalent continuous-time transfer function is

$$\frac{V_{S2}}{V_{slope}}(s) = \frac{\lambda \cdot e^{-0.5 \cdot sT}}{1 - (1 - \lambda) \cdot e^{-sT}} \cdot \frac{1 - e^{sT}}{sT}, \quad \text{in which } T = 1/f_{ref}. \quad (6)$$

Thus, the complete transfer function for the SPD shown in Fig. 12(a) can be calculated as

$$K_{SPD}(s) = K_{SPD} \cdot \frac{V_{S2}}{V_{slope}}(s) \cdot \frac{R_2 C_2 \cdot s}{1 + R_2 C_2 \cdot s}. \quad (7)$$

Note that the IIR filter introduces a high-frequency pole in the loop that reduces the phase margin. In this design, the pole is at $f_{ref}/4\pi = 8$ MHz, as $C_{S1} = C_{S2}$. Sampling capacitors of 200 fF are used in this design to keep the sampler PN floor below -170 dBc/Hz. Thick-oxide NMOS switch is used for sampling to reduce the switch ON-resistance variation over input voltage ramp. The sampling clocks CK1 and CK2 are level-shifted up to 1.8 V.

As explained in Section II, the noise, accuracy, and the input referred offset requirements of the G_M circuit are all much relaxed compared to a PFD-CP PLL, due to the architectural advantages of the proposed PLL. A single-ended G_M [see Fig. 12(a)] is used in this design and it consumes only a few microamperes current. The integration capacitor, C_1 , required for the optimal IPN is ~ 30 pF, corresponding to the PLL BW of ~ 600 kHz.

The schematic of the 6-GHz VCO is shown in Fig. 12(c). The VCO employs a thick-oxide complementary cross-coupled pair with tail inductor degeneration in order to achieve low PN (-125 dBc/Hz at 1-MHz offset) [24]. To track the wide temperature drift possible when integrated in a millimeter-wave phase array transceiver, a separate varactor of K_{VCO_T} , tuned by a proportional to absolute temperature (PTAT) voltage, is used to compensate the VCO frequency drift over temperature.

VI. EXPERIMENTAL RESULTS

This analog sampling PLL is implemented in a 28-nm bulk CMOS and is incorporated in a 28-GHz cellular RFIC. The PLL locks across the VCO tuning range, from 5.5 to 7.3 GHz

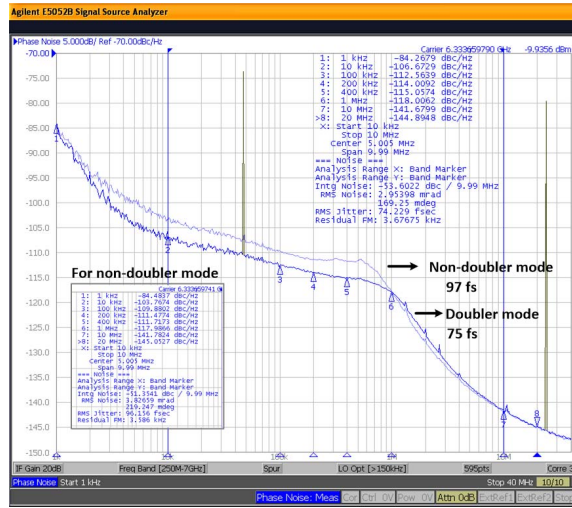
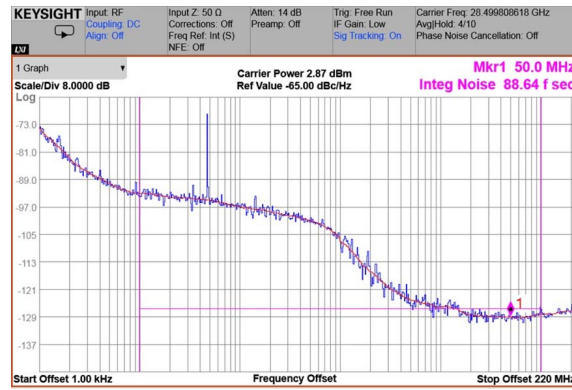
Fig. 13. PLL PN at a fractional- N channel with/without doubler.

Fig. 14. PLL PN measured at 28-GHz TX output.

and is multiplied up to provide the desired LO frequencies. The PLL performance is measured at the PLL output as well as at the 28-GHz transmitter (TX) output. Fig. 13 shows the measured PN spectrum at the PLL output for both non-doubler ($f_{\text{ref}} = 52$ MHz) and doubler modes ($f_{\text{ref}} = 104$ MHz). The in-band (at 400 kHz) PN is as low as -115 dBc/Hz in doubler mode for a fractional- N channel of 6.33 GHz, which is 3 dB lower than the non-doubler case. At the 10-kHz frequency offset, 5 dB improvement is observed with doubler. Thanks to the background DCC, the rms jitter of the PLL is reduced from 96.2 fs (-51.4 dBc) to 74.2 fs (-53.6 dBc), when integrated from 10 kHz to 10 MHz. Overall ~ 2 -dB IPN improvement is achieved in doubler mode, which agrees with the linear analysis in Fig. 2. The PLL test output buffer starts to contribute significant amount of PN from 10-MHz frequency offset onward. Fig. 14 depicts the correspondent PN spectrum measured at the TX output. The rms jitter is 88.64 fs (-39 dBc) at the carrier frequency of 28.5 GHz, when integrated from 10 kHz to 200 MHz, which only increases 14.4 fs compared to the rms jitter of the 6-GHz PLL.

Measured DTC resolution is 642 fs and the duty cycle error is ~ 48.5 ps according to the calibration results. The measured reference spur is -70.06 dBc from the PLL output spectrum

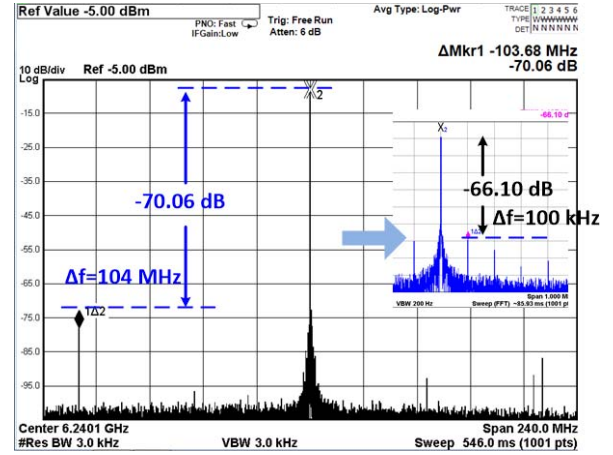


Fig. 15. Measured PLL output spectrum at 6240.1 MHz.

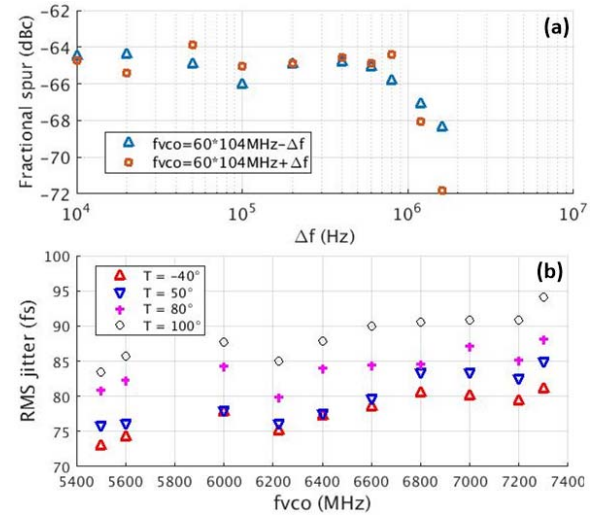


Fig. 16. (a) Measured fractional spur across the fractional codes. (b) PLL rms jitter across the tuning range over temperature.

in Fig. 15, which degrades the rms jitter by < 1 fs. The close-in spectrum of a near integer- N channel, 6240.1 MHz (104 MHz $\times 60 + 100$ kHz), is also shown in Fig. 15 with fractional spur of -66.1 dBc at 100-kHz offset. The measured fractional spur across fractional codes are shown in Fig. 16(a) for MASH 1-1 mode, which is close to the spur level predicted by system simulation using the DTC INL curve shown in Fig. 11. The worst case spur is -64 dBc without DTC nonlinearity calibration, which compares favorably to the design that used nonlinearity calibration [6], [26]. The PLL rms jitter across the tuning range is shown in Fig. 16(b), which degrades only ~ 20 fs from -40°C to 100°C with the help of the background calibrations. It also indicates a good DTC linearity over temperature variation.

The complete PLL draws 18.9 mW, in which VCO and DTC draw 13.5 and 2.7 mW, respectively, from a 1.8-V supply. The MMDIV consumes 1.1 mW. The sampling PD, G_M , and the calibration circuitries together draw 1.6 mW. Fig. 17 shows the chip photomicrograph. The PLL, excluding the routing to the frequency multiplier, occupies 0.45 mm 2 .

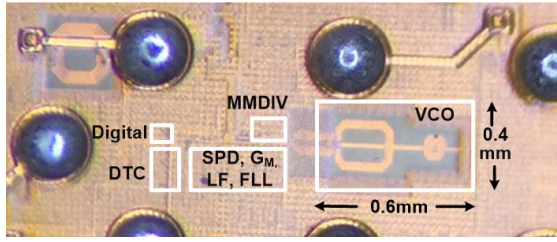


Fig. 17. Chip photomicrograph.

TABLE I

PERFORMANCE SUMMARY AND COMPARISON TO THE PRIOR ART

	JSSC' 2016 [6]	ISSCC' 2016 [8]	JSSC' 2017 [2]	ISSCC' 2014 [22]	This work
Technology	28nm	28nm	14nm	40nm	28nm
Type	Sub-sampling Analog	Sampling Digital	TDC Digital	PFD-CP Analog	Sampling Analog
f_{ref} (MHz)	40	40	26	26x2	52x2
f_{out} (GHz)	11.72	5.825	2.69	3.88	6.33
rms jitter (fs)	176	159	137	190	75
In-band PN,* $\mathcal{L}_{norm}$ (dBc/Hz)	-113.3	-104.8	-106.2	-101.2	-115.1
Frac. spur (dBc)	<-56.6	<-54	<-78.6	<-65	<-64
Ref. spur (dBc)	-69	-78	-87.6	-100	-70.2
Power (mW)	5.6	8.2	13.4	17.5	18.9
FoM (dB)	-247.6	-246.8	-246	-242	-249.7
Area (mm ²)	0.25	0.3	0.257	0.29	0.45

$$FoM = 10 \cdot \log_{10}((\sigma / 1 \text{ s})^2 \cdot \text{Power} / 1 \text{ mW}) \quad [27]$$

$$*\text{Normalized to 6.33 GHz, } \mathcal{L}_{norm}\{\Delta f\} = \mathcal{L}\{\Delta f\} + 20 \cdot \log_{10}(f_{out}/6.33 \text{ GHz})$$

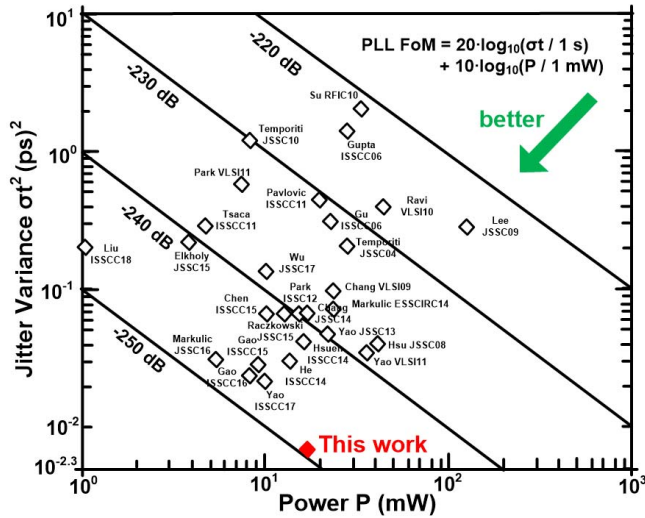


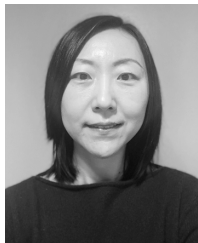
Fig. 18. Jitter and power comparison between this paper and prior art.

Performance summary and comparison are shown in Table I. This PLL achieves much lower rms jitter (75 fs), lower in-band PN, and better FoM (−249.7 dB) comparing to the prior art. The measured spurious tone performance is also comparable. Fig. 18 shows the jitter and power performance of this paper and a few state-of-the-art analog and digital PLLs. Although this PLL consumes 18.9 mW, it achieves lower jitter and better PLL FoM. This paper meets the stringent requirements of the 5G NR operating in the millimeter-waveband, and it is also an excellent candidate to be used for other high-performance applications.

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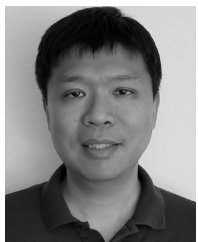
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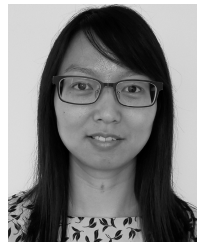
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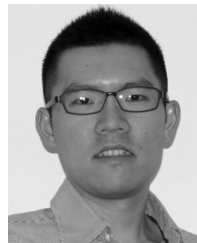


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In 2011, she was an Intern with RF Group, Analog Devices, Wilmington, MA, USA, where she was designed millimeter-wave quadrature modulator. Since 2013, she has been with Samsung Semiconductor Inc., San Jose, CA, USA, where she is currently a Staff Design Engineer, and developing

RF transceivers for cellular applications.

Dr. Ni was a recipient of the Analog Devices Graduate Fellowship in 2011.



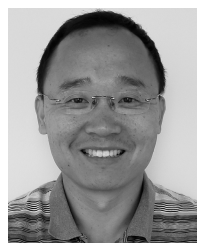
Pei-Yuan Chiang (S'12) received the B.S. and M.S. degrees in electrical and computer engineering from National Chiao Tung University, Hsinchu, Taiwan, in 2006 and 2008, respectively, and the Ph.D. degree in electrical engineering from the University of California at Irvine, Irvine, CA, USA, in 2014.

From 2014 to 2016, he was a Senior RFIC Design Engineer with Marvell Semiconductor Inc., Santa Clara, CA, USA, where he was working on Wifi/IOT circuitry. He is currently with Samsung Semiconductor Modem Research and Development Group, San Jose, CA, USA, where he is involved in advanced cellular RFIC design. His research interests include millimeter-wave and terahertz-integrated circuit design for wireless and high-speed data communication.



Yongping Han was born in Shanghai, China. She received the Ph.D. degree in electrical engineering from Arizona State University, Tempe, AZ, USA, in 2012.

She was with Texas Instruments, Dallas, TX, USA, where she was involved in the development of delta-sigma ADC and RFMD for CMOS power amplifier development. In 2014, she joined Samsung Semiconductor, San Jose, CA, USA. Her research interests are high-frequency circuit design for cellular PLL and 5G millimeter-wave application.



Yongrong Zuo received the B.S. and M.S. degrees in electronic engineering from Tsinghua University, Beijing, China, in 1994 and 1997, respectively, and the Ph.D. degree in electrical engineering from the University of Delaware, Newark, DE, USA, in 2006.

From 2006 to 2013, he was a Senior Engineer with Qualcomm, Boxborough, MA, USA, where he was a Staff Engineer of RF and analog IC design for various wireless applications. Since 2013, he has been a Senior Staff Engineer with Samsung Semiconductor Inc., San Jose, CA, USA, where he was designing low-power and low-voltage RF and analog ICs using advanced deep-submicrometer CMOS technologies.



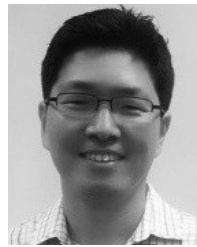
Ashutosh Verma received the B.Tech. degree in electrical engineering from the IIT Delhi, New Delhi, India, in 1999, and the M.S.E.E. and Ph.D.E.E. degrees from the University of California at Los Angeles, Los Angeles, CA, USA, in 2004 and 2009, respectively.

From 2000 to 2002, he was with Biomorphic VLSI Inc., Thousand Oaks, CA, USA, where he was designed digital-integrated circuits for CMOS imagers. From 2009 to 2015, he was with Marvell Semiconductor, Santa Clara, CA, USA, where he designed mixed-signal and baseband circuits for WLAN transceivers. Since 2015, he has been with Samsung Semiconductor, San Jose, CA, USA, where he was working on circuits for cellular transceivers.



Ivan Siu-Chuang Lu (M'04) was born in Taipei, Taiwan, in 1979. He received the B.E.E. degree from the University of Queensland, Brisbane, Australia, in 2001, and the Ph.D. degree from the University of New South Wales, Sydney, Australia, in 2006.

He was with Marvell Semiconductor, Santa Clara, CA, USA, and with Mediatek, San Jose, CA, USA, where he lead and contributed to transceiver designs for cellular, multi-in multi-out (MIMO) WiFi, and GPS applications. Since 2014, he has been with Samsung Semiconductor, San Jose, CA, USA, where he is currently a Director of Advance Circuit Design Division. He holds 13 issued and several pending patents in the field of RF and analog circuit design. His current research interests include advanced RF CMOS transceiver architectures and circuits for wireless communication system.



Sang Won Son (S'96–M'02) received the B.S., M.S., and Ph.D. degrees in electrical engineering and computer science from the University of California at Berkeley, Berkeley, CA, USA, in 1995, 1998, and 2002, respectively.

From 2002 to 2010, he was with Marvell Technology Group, Santa Clara, CA, USA, where he worked on multiple generations of WLAN transceiver products. From 2010 to 2012, he was a Senior Department Manager with Mediatek, San Jose, CA, USA, where he was developed RF and analog circuits for cellular and connectivity applications. He is currently a Vice President with Samsung Semiconductor, Inc., San Jose, CA, USA, where he is involved in developing the next generation cellular chipset solutions.



Thomas Byunghak Cho (SM'14) received the B.S. degree in electrical engineering from the University of California at Los Angeles (UCLA), Los Angeles, CA, USA, and the M.S. and Ph.D. degrees in electrical engineering from the University of California at Berkeley, Berkeley, CA, USA.

In 1996, he joined Level One Communications, where he was involved in developing CMOS RF transceiver products for cordless phone applications. In 2000, he co-founded Wireless Interface Technologies, where he was involved in developing CMOS RF transceivers for WPAN/WLAN applications. In 2004, he joined Marvell Semiconductor, Santa Clara, CA, USA, where he was involved in developing CMOS RF and analog IC products for various wired and wireless connectivity applications. Since 2012, he has been with Samsung Electronics, Hwaseong-si, South Korea, where he is currently the Executive Vice President of Infra and Design Technology Center in charge of cellular/connectivity RF transceiver development, digital physical implementation, and design verification.